

ADAM TAY XUE HAO

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 1.5 years of experience in developing LLM at the edge solutions for enterprise use cases and LLM fine-tuning. Skilled in RAG, Text-to-SQL, Workflows and Model enablement, with hands-on experience in optimizing AI performance on Intel NPU. Published a paper on Deep Reinforcement Learning as the main author in a Scopus Q2 journal and recognized for delivering impactful enterprise AI solutions featured at CES 2025 and MWC 2025.

WORKING EXPERIENCE

❖ **System Engineer in AI, Intel Corporation** Jan 2024 — Present
Penang

Intel AI Assistant Builder:

- Core AI developer of an **edge AI solution**, building a fully **local LLM** application using **LlamaIndex** in **Python**, exposing features through **gRPC** API.
- Extended Huggingface's Transformers library to perform multi-token zero-shot classification, achieving more efficient and predictable workflows; Work submitted for **patent filing**.
- Developed LLM features including RAG, Text-to-SQL, Summarization, and Resume Scoring workflows, along with agent functionalities.
- Enabled Multimodal Large Language Models using Optimum intel for image querying and generation.
- Developed and maintained a CICD pipeline using GitHub Action for automated building and testing.

NPU Kernel Optimization with Boolean Packing:

- Optimized **memory footprint** of the Select kernel on NPU by **representing Booleans in INT1** using C++.
- Achieved **153% performance boost** in ResNet50 performance on Intel NPU.

LLM Code Generation Fine-tuning:

- Performed PEFT (Parameter Efficient Fine-tuning) using **LoRA** with **Unsloth** to fine-tune domain specific code (Device Modeling Language) generation.
- Achieved **84.8%** accuracy after finetuning using **LLM-as-a-judge + Domain Expert Evaluation**. Accuracy of the **base model** was **36.1%**,

Initiatives:

- Mentored 2 junior engineers to build custom chunking solutions for HTML documents to reduce chunk fragmentation, leading to **37.5%** overall improvements in accuracy using **LLM-as-a-judge** with GPT-4o.
- Leading a team of 3 AI engineers in the ongoing PEFT fine-tuning for SQL using Supervised Finetuning and Reinforcement Finetuning with GRPO.
- Initiated a learning group consisting of 3 junior engineers to study model graph partitioning techniques using the Openvino NPUW framework to enable day 0 LLM support on Intel NPU.

❖ **Freelance AI Engineer, Fiverr** Aug 2023 — Dec 2023
Remote

- **Secured 3 repeat orders from a client:**
 - Performed PEFT using QLoRA to fine-tune Falcon 7b on mental health datasets.
 - Conducted data transformation for fine-tuning by converting raw text into JSON format with prompt-response pairs.
 - Applied semi-supervised learning with self-training for MBTI personality prediction on a 20:80 labeled-to-unlabeled dataset of Twitter comments. Achieved **0.91 average f1-score across 4 categories**.

EDUCATION

❖ **Tunku Abdul Rahman University of Management and Technology** May 2021 — Aug 2023
Bachelor of Computer Science (Honours), Data Science *Kuala Lumpur*

- Grade: **4.0/4.0 CGPA**
- Book Prize Award Winner | First Class Honours

PUBLICATIONS AND ACHIEVEMENTS

- ❖ **Publication**
 - Main author. Paper accepted by a Scopus Q2 Journal.
 - Tay, X., H. & Lim, S., M. (2024, Sep). **Deep Reinforcement Learning in Cryptocurrency Trading: A Profitable Approach**, Journal of Telecommunications and the Digital Economy, Vol. 12 No. 3, 126-147
- ❖ **Achievement: Intel AI Assistant Builder**
 - Delivered a demo-ready LLM RAG application within 7 weeks for **Computex 2024**.
 - Application featured at **CES 2025**.
 - Application won the **Best AI Debut** at **MWC 2025**.

TECHNICAL SKILLS

- ❖ **Core Skills**
Decoder-only Architectures, Local LLM, RAG, Fine-tuning, Agents, Model Optimization
- ❖ **Programming Languages**
Python, SQL, C++, Java
- ❖ **Fundamentals**
GitHub Actions, Docker, Bash scripting, gRPC, SOLID/DRY, Data Science, Machine Learning, Backend programming
- ❖ **Natural Language**
English, Mandarin, Malay

REFERENCES

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